

Final_Presentation_Gabriel

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```
#install.packages('tidyverse')  
library(tidyverse) # For data wrangling
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.0      v readr      2.2.0  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2    4.0.2      v tibble     3.3.1  
v lubridate  1.9.5      v tidyr      1.3.2  
v purrr      1.2.1  
-- Conflicts ----- tidyverse_conflicts() --  
x dplyr::filter() masks stats::filter()  
x dplyr::lag()     masks stats::lag()  
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
#install.packages('brms')  
library(brms) # For stats
```

Loading required package: Rcpp
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing `help('brms')`. A more detailed introduction
to the package is available through `vignette('brms_overview')`.

Attaching package: 'brms'

The following object is masked from 'package:stats':

ar

```
#install.packages('ggeffects')
library(ggeffects) # for plotting model predictions
# Note: I needed to also install the `insight` and `see` packages to get `modelbased` to ins
#install.packages('insight')
library(insight)
#install.packages('see')
library(see)
#install.packages('modelbased') # if you need to install this package
library(modelbased) # for plotting model predictions. supports the link scale (ggeffects doe
```

Attaching package: 'modelbased'

The following objects are masked from 'package:ggeffects':

collapse_by_group, pool_predictions, residualize_over_grid

```
#install.packages('faraway') # if you need to install this package
library(faraway) # For data on galapagos species richness
```

Attaching package: 'faraway'

The following object is masked from 'package:brms':

epilepsy

```
#install.packages('ggplot2')
library(ggplot2)
#install.packages('here')
library(here)
```

here() starts at /Users/gabegott/School/ds4eeb/ds4eeb_Final_Project

```
#install.packages('dagitty') # only the first time
library(dagitty)
```

Data Wrangling

First we are going to combine the data into one data frame. This will make using the data later on easier as we will only have to refer to one data frame when creating plots and models. Left join will be great for this as it will ensure that all data ends up in the proper location. There were some issues with the way the data moved when done at once, so it has been broken down into several steps. Luckily, we will not have to pivot any data as it is already in the proper format.

```
#This chunk is intended to combine the no3, climind df, and temp data frame
#Read the CSV files into R
kelp_data <- read.csv("rawdata/kelp_no3_waves_quarterly_long.csv")
clim_data <- read.csv("rawdata/climind_quarterly.csv")

#Combine the data frames
combined_sitespecific_climind <- left_join(kelp_data, clim_data, by = c("year", "quarter"))

#View the new dataframe
head(combined_sitespecific_climind)
```

	site_id	year	quarter	kelp	no3	waves	NPGO	MEI	PDO
1	1	1987	1	0	5.08	3.638	0.58	1.2	1.65
2	2	1987	1	23572	5.29	6.818	0.58	1.2	1.65
3	3	1987	1	771	4.62	4.519	0.58	1.2	1.65
4	4	1987	1	1856	4.71	4.400	0.58	1.2	1.65
5	5	1987	1	0	4.43	4.351	0.58	1.2	1.65
6	6	1987	1	10423	4.42	4.391	0.58	1.2	1.65

```
#Save the combined data frame as a new CSV file
write.csv(combined_sitespecific_climind, file = "newdf/combined_kelp_climind_data.csv", row.names = FALSE)

#Read the temperature data
temp_data <- read.csv("rawdata/SBH_temperature_20241231.csv")

#Convert daily data to quarterly averages
temp_quarterly <- temp_data %>%
  # Create a "quarter" column based on the MONTH (e.g., Jan-Mar = 1, Apr-Jun = 2, etc.)
  mutate(quarter = ceiling(MONTH / 3)) %>%
  # Rename YEAR to lowercase "year" so it matches your other datasets
  rename(year = YEAR) %>%
  # Group by the year and our newly created quarter
  group_by(year, quarter) %>%
```

```
# Calculate the average surface temperature for that quarter (ignoring missing data)
summarize(mean_surf_temp = mean(SURF_TEMP_C, na.rm = TRUE), .groups = "drop")

#Join the new quarterly temperature data to your combined dataframe
combined_sitespecific_climind_temp <- left_join(combined_sitespecific_climind, temp_quarterly)

#View the results
head(combined_sitespecific_climind_temp)
```

	site_id	year	quarter	kelp	no3	waves	NPGO	MEI	PDO	mean_surf_temp
1	1	1987	1	0	5.08	3.638	0.58	1.2	1.65	14.108
2	2	1987	1	23572	5.29	6.818	0.58	1.2	1.65	14.108
3	3	1987	1	771	4.62	4.519	0.58	1.2	1.65	14.108
4	4	1987	1	1856	4.71	4.400	0.58	1.2	1.65	14.108
5	5	1987	1	0	4.43	4.351	0.58	1.2	1.65	14.108
6	6	1987	1	10423	4.42	4.391	0.58	1.2	1.65	14.108

```
#save df as file
write.csv(combined_sitespecific_climind_temp, file = "newdf/combined_sitespecific_climind_temp.csv")
combined_data = read.csv(here("newdf/combined_sitespecific_climind_temp.csv"))
```

Data Searching

We are going to plot several different relationships to see how they may correlate. This alone will not give us enough information to draw conclusions, but it will help us learn where to start looking. The plots created are those which determined by the DAG have a direct impact on kelp biomass

Each relationship will be demonstrated on two separate plots. In the first plot the data will be modeled by both `geom_smooth` model “lm” and `geom_smooth` model “gam” (“gam” is used instead of “loess” as loess would not properly display. We believe that this is due to the sheer number of samples). When looking at the plot we can see how well the “gam” model fits with the “lm” model. If they are close together then the data is likely linear, if they deviate strongly then it is likely non-linear. The second plot for each relationship will be on a log scale. This is done to see if the relationship is logarithmic. If the slope of the data appears to become linear in these models then we know that the relationship is logarithmic.

NO3 vs Kelp Biomass

Here we run the relationships for nitrate Vs Kelp biomass.

```

#Plotting no3 vs biomass to find if the data is linear or nonlinear
p.nitrateVsbiomass.linearitycheck = ggplot(combined_data, aes(x = no3, y = kelp)) +
  stat_summary_bin(
    bins = 60,
    fun = mean,
    geom = "point"
  )+
  labs(
    title = "Relationship Between Nitrate Concentrations and Kelp Biomass",
    x = "Nitrate Concentration (µmol/L)",
    y = "Kelp Biomass (Kg)"
  )+
  theme_classic()+
  annotate(
    "text",
    x = Inf, y = Inf,
    label = paste("Slope =", round(coef(lm(kelp ~ no3, data = combined_data))[2], 3)),
    hjust = 1.1, vjust = 1.5
  )+
  geom_smooth(method = "lm", se = FALSE, color = "red", linewidth = 1.2)+
  geom_smooth(method = "gam", formula = y ~ s(x) , se = TRUE, linetype = "dashed", color = "red")

ggsave(
  filename = here("outputs", "p.nitrateVsbiomass.linearitycheck.png"),
  plot = p.nitrateVsbiomass.linearitycheck
)

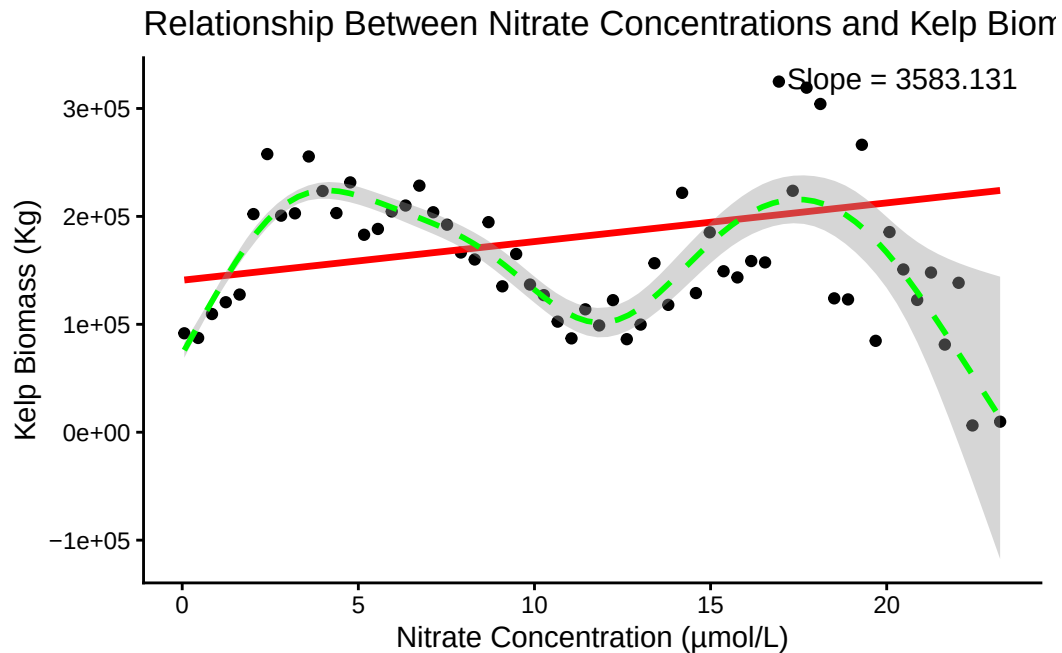
```

Saving 5.5 x 3.5 in image

`geom\_smooth()` using formula = 'y ~ x'

```
p.nitrateVsbiomass.linearitycheck
```

`geom\_smooth()` using formula = 'y ~ x'



```
#Plotting nitrate vs biomass against a log scale
p.nitrateVsbioass.logscale = ggplot(combined_data, aes(x = no3, y = kelp)) +
  stat_summary_bin(
    bins = 60,
    fun = mean,
    geom = "point"
  ) +
  scale_x_log10() +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  theme_classic() +
  labs(
    title = "Kelp Biomass vs Nitrate Concentration",
    x = "Nitrate Concentration (log10 scale)",
    y = "Kelp Biomass (kg)"
  )

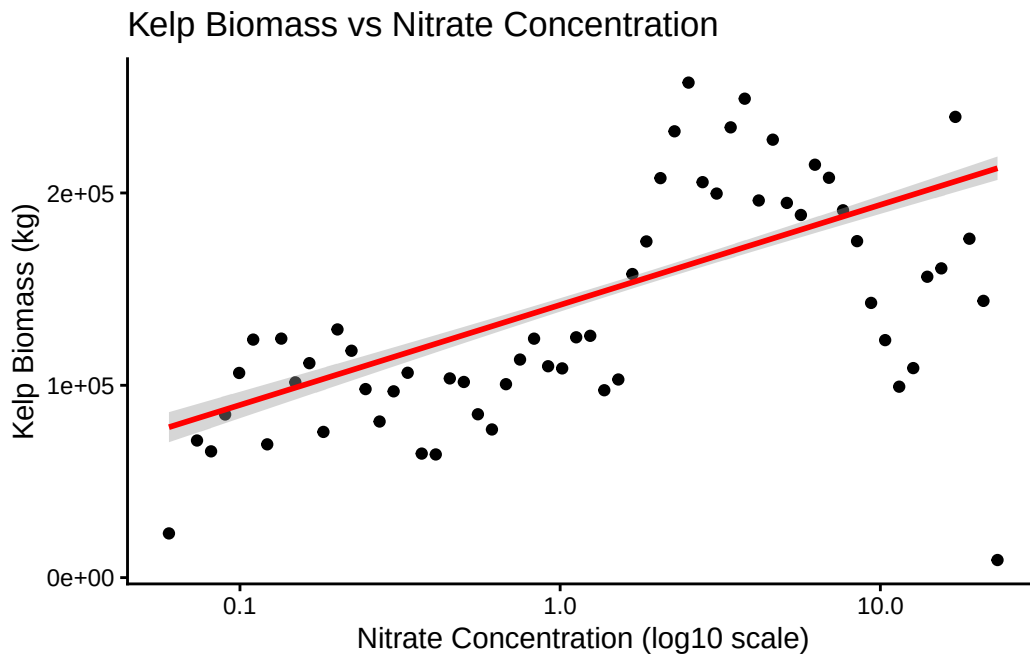
ggsave(
  filename = here("outputs", "p.nitrateVsbioass.logscale.png"),
  plot = p.nitrateVsbioass.logscale
)
```

Saving 5.5 x 3.5 in image

`geom\_smooth()` using formula = 'y ~ x'

```
p.nitrateVsbiomass.logscale
```

```
`geom_smooth()` using formula = 'y ~ x'
```



We can see that the biomass certainly changes as the nitrate concentration changes. On the first plot we see that the data takes more of an M shape. On the second plot we do see that the data begins to make a more linear path. For this reason the relationship between nitrate and Vs kelp biomass is approximately logarithmic.

Wave Height Vs. Kelp Biomass

Here we are creating plots to model wave height Vs. biomass

```
#Creates a scatter plot to show the relationship between waves and biomass  
  
p.wavesVsbiomass = ggplot(combined_data, aes(x = waves, y = kelp)) +  
  stat_summary_bin(  
    bins = 60,  
    fun = mean,  
    geom = "point"  
  )+
```

```

geom_smooth(method = "lm", color = "red") +
annotate(
  "text",
  x = Inf, y = Inf,
  label = paste("Slope =", round(coef(lm(kelp ~ waves, data = combined_data))[2], 3)),
  hjust = 1.1, vjust = 1.5
)+
  geom_smooth(method = "gam", formula = y ~ s(x) , se = TRUE, linetype = "dashed", color =
labs(
  title = "Relationship Between Wave Height and Kelp Biomass",
  x = "Wave Height (M)",
  y = "Kelp Biomass (Kg)"
) +
theme_classic()

ggsave(
  filename = here("outputs", "p.wavesVsbiomass.png"),
  plot = p.wavesVsbiomass
)

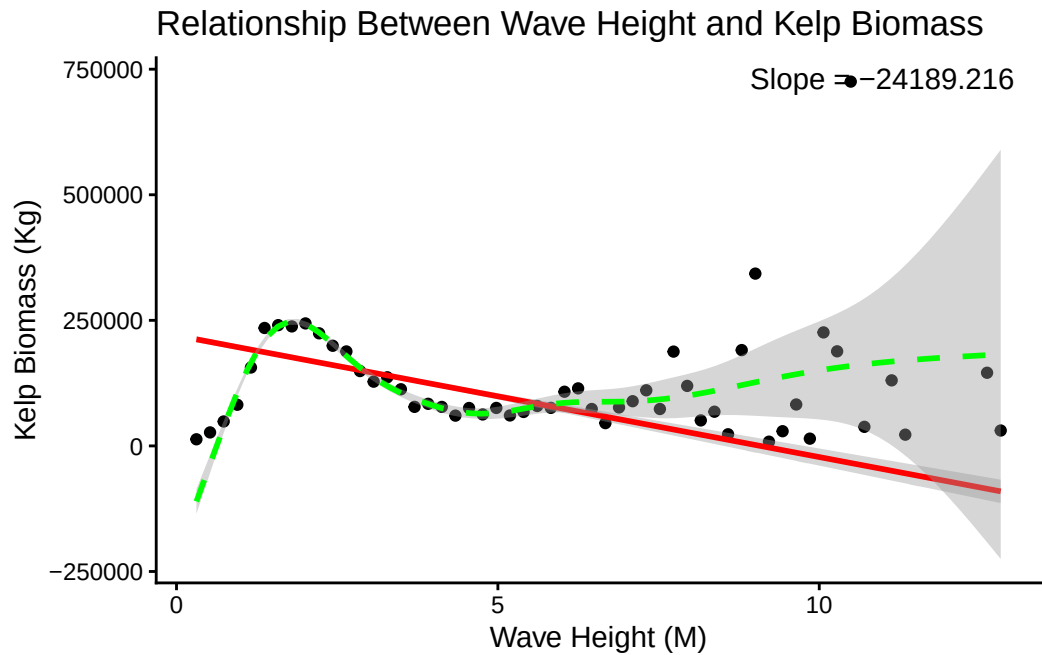
```

Saving 5.5 x 3.5 in image

`geom\_smooth()` using formula = 'y ~ x'

p.wavesVsbiomass

`geom\_smooth()` using formula = 'y ~ x'



```
#Creates a scatter plot to show the relationship between waves and biomass with a log scale

p.wavesVsbiomass.logarithmic = ggplot(combined_data, aes(x = waves, y = kelp)) +
  stat_summary_bin(
    bins = 60,
    fun = mean,
    geom = "point"
  ) +
  geom_smooth(method = "lm", color = "red") +
  annotate(
    "text",
    x = Inf, y = Inf,
    label = paste("Slope =", round(coef(lm(kelp ~ waves, data = combined_data))[2], 3)),
    hjust = 1.1, vjust = 1.5
  ) +
  scale_x_log10() +
  geom_smooth(method = "gam", formula = y ~ s(x), se = TRUE, linetype = "dashed", color =
  labs(
    title = "Relationship Between Wave Height and Kelp Biomass",
    x = "Wave Height (M)",
    y = "Kelp Biomass (Kg)"
  ) +
  theme_classic()
```

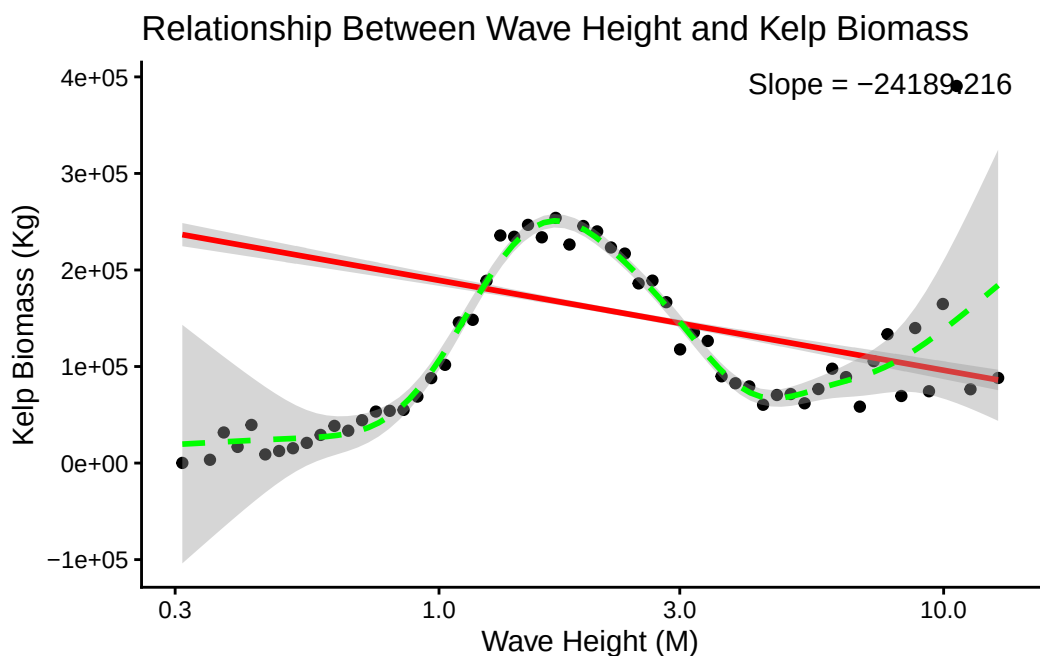
```
ggsave(
  filename = here("outputs", "p.wavesVsbiomass.logarithmic.png"),
  plot = p.wavesVsbiomass.logarithmic
)
```

Saving 5.5 x 3.5 in image

`geom\_smooth()` using formula = 'y ~ x'

p.wavesVsbiomass.logarithmic

`geom\_smooth()` using formula = 'y ~ x'



It doesn't appear that the pattern follows any family of patterns specifically. Looking at both models we can see that the data appears to peak in between 1 and 3 meters. Therefore wave height certainly does appear to have an impact on kelp biomass. When waves are too small ($<1\text{m}$) or too large (>3), the biomass of the kelp stays small. However, there is a sweet spot in between 1 and 3 meters. In this range the kelp is able to grow much more abundantly. There are several reasons that we can see this as a possibility. Perhaps at low levels of wave activity there is not enough nutrient flow in the water to promote growth. On the flip side, perhaps when the waves are too large they are breaking down the kelp, preventing growth. These are just a few thoughts that come to mind when looking at the graphs and will be better examined in the brms models later on.

Mean Surface Temperature Vs. Kelp Biomass

Here we examine the relationship of water temperature Vs kelp biomass.

```
p.tempVsbioass = ggplot(combined_data, aes(x = mean_surf_temp, y = kelp)) +
stat_summary_bin(
  bins = 60,
  fun = mean,
  geom = "point"
)+
geom_smooth(method = "lm") +
geom_smooth(method = "gam", formula = y ~ s(x) , se = TRUE, linetype = "dashed", color = "g")
annotate(
  "text",
  x = Inf, y = Inf,
  label = paste("Slope =", round(coef(lm(kelp ~ mean_surf_temp, data = combined_data))[2],
  hjust = 1.1, vjust = 1.5
)+
labs(
  title = "Relationship Between Mean Surface Temperature and Kelp Biomass",
  x = "Mean Surface Temperature (°C)",
  y = "Kelp Biomass (Kg)"
) +
theme_classic()

ggsave(
  filename = here("outputs", "p.tempVsbioass.png"),
  plot = p.tempVsbioass
)
```

Saving 5.5 x 3.5 in image

Warning: Removed 1805 rows containing non-finite outside the scale range
(`stat\_summary\_bin()`).

`geom\_smooth()` using formula = 'y ~ x'

Warning: Removed 1805 rows containing non-finite outside the scale range
(`stat\_smooth()`).

Warning: Removed 1805 rows containing non-finite outside the scale range
(`stat\_smooth()`).

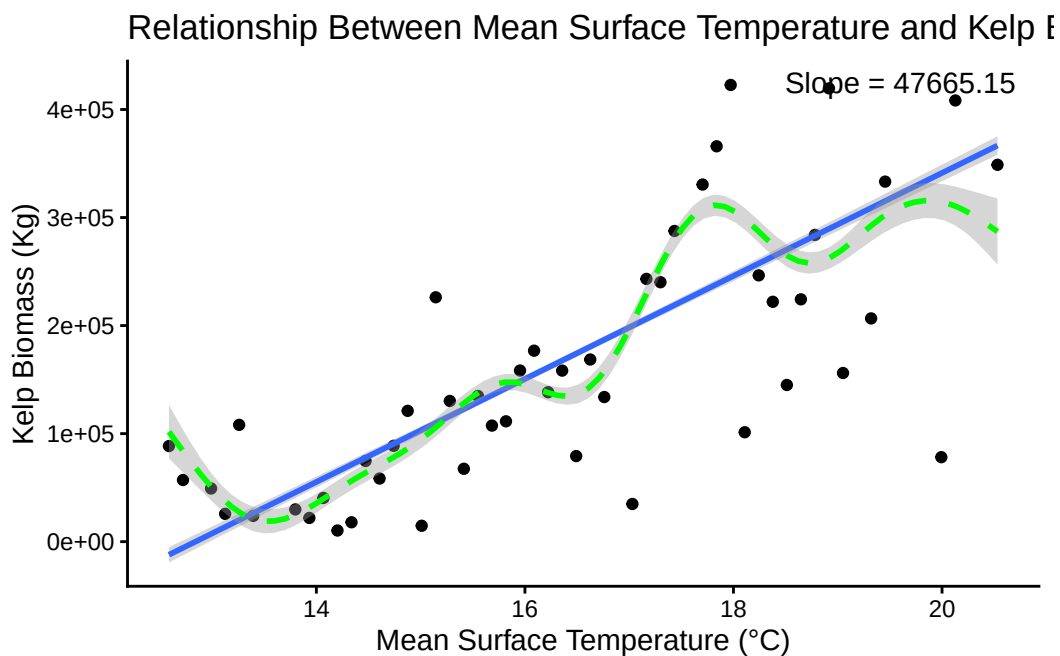
```
p.tempVsbiomass
```

```
Warning: Removed 1805 rows containing non-finite outside the scale range
(`stat_summary_bin()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1805 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Removed 1805 rows containing non-finite outside the scale range
(`stat_smooth()`).
```



```
#Plotting nitrate vs biomass against a log scale
#Not very informative because there is such a narrow temp range
p.tempVsbiomass.log = ggplot(combined_data, aes(x = mean_surf_temp, y = kelp)) +
  stat_summary_bin(
    bins = 60,
    fun = mean,
    geom = "point"
  ) +
  geom_smooth(method = "lm") +
```

```

annotate(
  "text",
  x = Inf, y = Inf,
  label = paste("Slope =", round(coef(lm(kelp ~ mean_surf_temp, data = combined_data))[2],
  hjust = 1.1, vjust = 1.5
)+
  scale_x_log10() +
labs(
  title = "Relationship Between Mean Surface Temperature and Kelp Biomass",
  x = "Mean Surface Temperature (°C)",
  y = "Kelp Biomass (Kg)"
) +
theme_classic()

ggsave(
  filename = here("outputs", "p.tempVsbiomass.log.png"),
  plot = p.tempVsbiomass.log
)

```

Saving 5.5 x 3.5 in image

Warning: Removed 1805 rows containing non-finite outside the scale range
(`stat\_summary\_bin()`).

`geom\_smooth()` using formula = 'y ~ x'

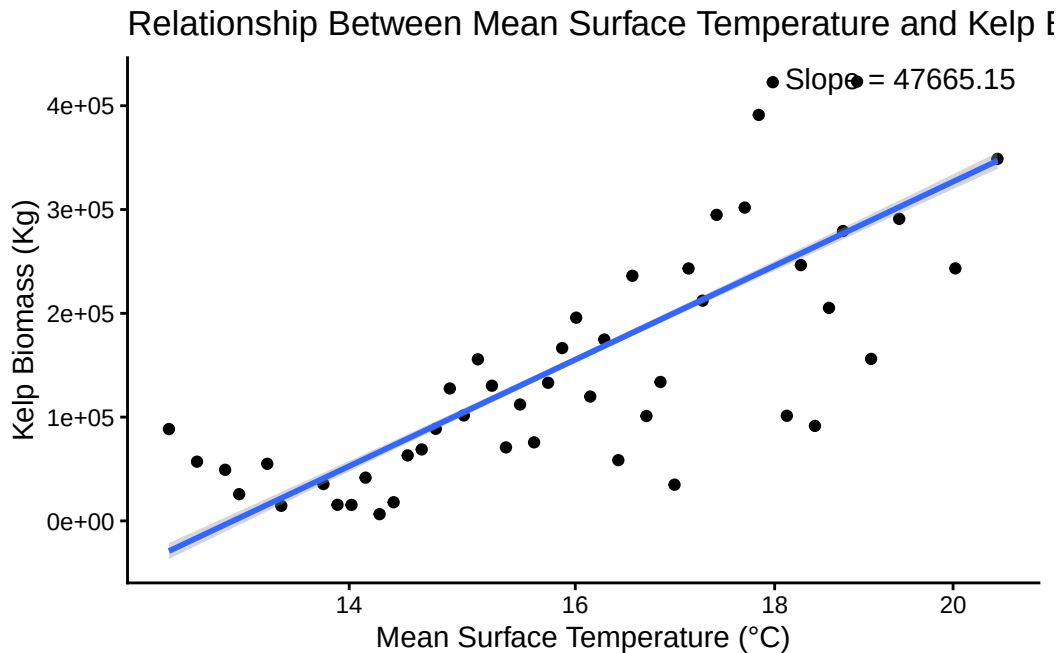
Warning: Removed 1805 rows containing non-finite outside the scale range
(`stat\_smooth()`).

p.tempVsbiomass.log

Warning: Removed 1805 rows containing non-finite outside the scale range
(`stat\_summary\_bin()`).

`geom\_smooth()` using formula = 'y ~ x'

Warning: Removed 1805 rows containing non-finite outside the scale range
(`stat\_smooth()`).



Here we see that the relationship between water temperature and biomass is clearly positive. However when comparing the two plots it isn't exactly clear whether or not it is linear or logarithmic. This is because when working with our temperature range, which is relatively narrow, the two plots look almost completely identical. It is our understanding we can use the simple linear model as our guide, as that appears to have a good fit on its own.

DAG

In our DAG our observed variables are; PDO (Pacific Decadal Oscillation), NPGO (North Pacific Gyre Oscillation), MEI (Multivariate ENSO Index), Temperature, Nitrate, Wave Height, and Kelp Biomass.

Our unobserved variables are Grazing and Eutrophication.

The observed variables' origins and units are explained in the rawdata README.

Pacific Decadal Oscillation (PDO):

A recurring pattern of water surface temperatures in the Pacific ocean north of 20°N changing between warmer and cooler temperatures every few decades. With positive values indicating a warming event, and negative values indicating a cooling event.

PDO causes change in temperature and nitrate concentration.

North Pacific Gyre Oscillation (NPGO):

A recurring phase of upwelling, in the form of counter rotating gyres, along 40°N in the Pacific.

When the NPGO is positive, upwelling-favorable conditions are seen in the California Current and Alaskan Gyre, while downwelling-favorable conditions are seen in the Subtropical Gyre and the Alaskan Coastal Current.

NPGO causes change in temperature and nitrate concentration through gyres with bring up deep waters.

Multivariate El Nino Southern Oscillation Index (MEI):

An index used to characterize the intensity of an El Nino event, with high positive values indicating a warming event, and high negative values indicating a cooling event.

MEI causes change in temperature, nitrate concentration, and wave height with eastern winds and storms.

Temperature

Measured in degrees C.

Temperature causes changes in nitrate concentration and kelp biomass by impacting metabolic rates of nitrate fixing organisms and kelp.

Nitrate

Measured in micro-moles per liter.

Nitrate affects kelp biomass because it is a nutrient required for plant growth.

Wave Height

Measured in meters.

Wave height affects nitrate concentration and kelp biomass.

Kelp Biomass

Measured in kilograms.

Kelp biomass is our outcome and if affected directly or indirectly by all other variables.

Grazing

An unobserved variable which would affect kelp biomass in the form of animals eating said kelp.

Eutrophication

An unobserved variable which could increase nitrate concentration in the form of fertilizer runoff.

```

kelp_dag <- dagitty('dag {
bb="-1.707,-4.687,2.5,4.206"
Eutrophication [latent,pos="0.778,0.628"]
Grazing [latent,pos="-0.694,1.288"]
Kelp_Biomass [outcome,pos="0.074,1.253"]
MEI [pos="-0.768,-0.889"]
NPGO [pos="0.392,-1.792"]
Nitrate [pos="0.079,-0.136"]
PDO [pos="-0.597,-1.734"]
Temperature [pos="0.812,-0.727"]
Wave_Height [pos="-0.738,0.535"]

Eutrophication -> Nitrate
Grazing -> Kelp_Biomass
MEI -> Nitrate
MEI -> Temperature
MEI -> Wave_Height
NPGO -> Nitrate
NPGO -> Temperature
Nitrate -> Kelp_Biomass
PDO -> Nitrate
PDO -> Temperature
Temperature -> Kelp_Biomass
Temperature -> Nitrate
Wave_Height -> Kelp_Biomass
Wave_Height -> Nitrate
}')

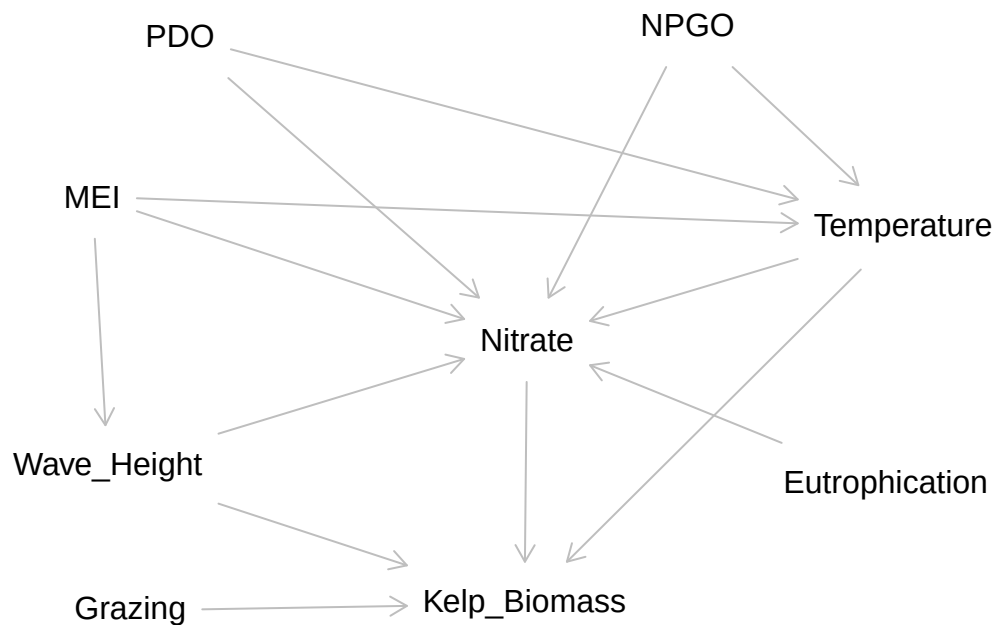
png(here("outputs", "kelp_dag.png"), width = 1600, height = 1200, res = 300)
plot(kelp_dag)
dev.off()

```

pdf

2

```
plot(kelp_dag)
```



```
impliedConditionalIndependencies(kelp_dag)
```

```

Kl_B _||_ MEI | Ntrt, Tmpr, Wv_H
Kl_B _||_ NPGO | Ntrt, Tmpr, Wv_H
Kl_B _||_ PDO | Ntrt, Tmpr, Wv_H
MEI _||_ NPGO
MEI _||_ PDO
NPGO _||_ PDO
NPGO _||_ Wv_H
PDO _||_ Wv_H
Tmpr _||_ Wv_H | MEI

```

BRMS Analysis

Now we can begin to analyze the relationships in models. For this we will create Bayesian models estimated using MCMC.

What variables are direct causes of nitrate concentrations according to our DAG?

Here we will analyze what variables have an impact on the nitrate concentration and how large of an affect they have.

```
#Including NPGO + PDO + MEI + mean_surf_temp + wave height
#estimating the direct effect of NPGO, PDO, Temperature, and MEI on Nitrate:
m.nitrate.MEI.NPGO.PDO.Temp.Waves <-
  brm(data = combined_sitespecific_climind_temp, # Give the model the combined data
    # Choose a gaussian (normal) distribution
    family = gaussian,
    # Specify the model here.
    no3 ~ NPGO + PDO + MEI + mean_surf_temp + waves,
    # Here's where you specify parameters for executing the Markov chains
    # We're using similar to the defaults, except we set cores to 4 so the analysis runs fa
    iter = 2000, warmup = 1000, chains = 4, cores = 4,
    # Setting the "seed" determines which random numbers will get sampled.
    # In this case, it makes the randomness of the Markov chain runs reproducible
    # (so that both of us get the exact same results when running the model)
    seed = 4,
    # Save the fitted model object as output - helpful for reloading in the output later
    file = "outputs/m.nitrate.MEI.NPGO.PDO.Temp.Waves")

summary(m.nitrate.MEI.NPGO.PDO.Temp.Waves)
```

```
Family: gaussian
Links: mu = identity
Formula: no3 ~ NPGO + PDO + MEI + mean_surf_temp + waves
Data: combined_sitespecific_climind_temp (Number of observations: 45847)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Regression Coefficients:

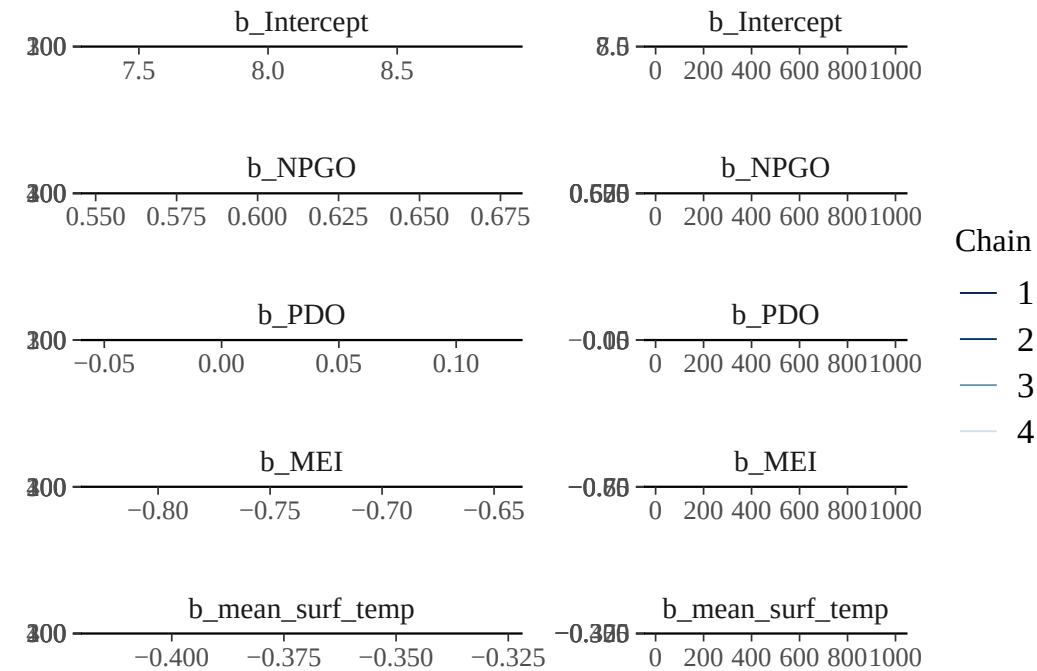
	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	8.19	0.21	7.78	8.59	1.00	4723	2834
NPGO	0.61	0.02	0.57	0.64	1.00	5322	2813
PDO	0.04	0.03	-0.01	0.09	1.00	4190	3448
MEI	-0.73	0.02	-0.77	-0.68	1.00	4247	3107
mean_surf_temp	-0.38	0.01	-0.40	-0.35	1.00	5156	2984
waves	0.78	0.01	0.75	0.81	1.00	4877	3082

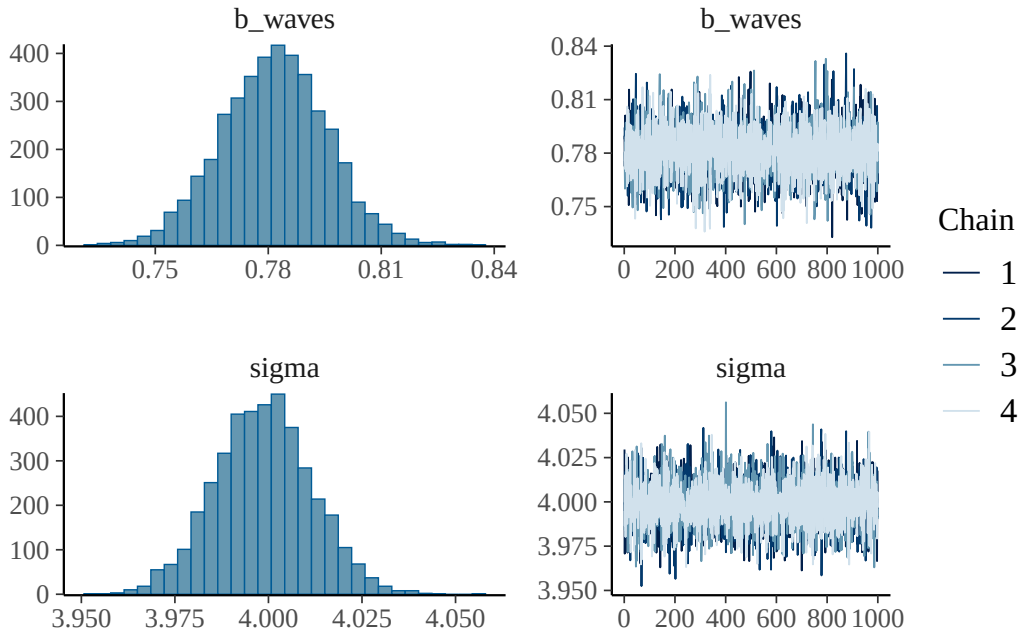
Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	4.00	0.01	3.97	4.02	1.00	6324	3135

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(m.nitrate.MEI.NPGO.PDO.Temp.Waves)
```





Now we can analyze what factors have a direct impact on the nitrate levels.

NPGO: It appears that for every one increase in the NPGO (unit-less index) then the nitrate levels increase by 0.61 micromoles per liter. The effect was significantly different from zero because the compatibility interval ranged from 0.57 to 0.64 micromoles per liter.

PDO: PDO did not have an impact on nitrate levels that was significantly different from zero. This is because the compatibility interval ran from -0.01 to 0.09 micromoles per liter.

MEI: For every one increase in MEI (unit-less index) the nitrate level decreases by 0.73 micromoles per liter. The change was significantly different from zero because the compatibility interval ran from -0.77 to -0.68 micromoles per liter.

Water Temperature: For every 1 °C increase in water temperature the nitrate level decreased by -0.38. The affect was significantly different from zero because the comatibility interval ran from -0.40 to -0.35 micromoles per liter.

Waves: For every one meter increase in wave height the nitrate levels increased at an average of 0.78 micromoles per liter. The affect was significantly different from zero because the comatibility interval ran from 0.75 to 0.81 micromoles per liter.

Now these points alone do not tell us what impact these have on kelp biomass, but they do show us the direct effect that they have on nitrate level. This is useful because we will be using nitrate level as a part of out kelp biomass model.

What variables are direct causes of mean surface temperature according to our DAG?

Here we will analyze what variables have an impact on the surface water temperature and how large of an affect they have.

```
#Including MEI, NPGO and PDO to see their affect on temp
m.temp.MEI.NPGO.PDO <-
  brm(
    data = combined_sitespecific_climind_temp, # Give the model the combined data
    family = gaussian, # Choose a gaussian (normal) distribution
    mean_surf_temp ~ MEI + NPGO + PDO, #using MEI NPGO and PDO to analyze temp
    iter = 2000, warmup = 1000, chains = 4, cores = 4,
    seed = 4, #which random numbers will get sampled
    file = "outputs/m.temp.MEI.NPGO.PDO"
  )

summary(m.temp.MEI.NPGO.PDO)
```

```
Family: gaussian
Links: mu = identity
Formula: mean_surf_temp ~ MEI + NPGO + PDO
Data: combined_sitespecific_climind_temp (Number of observations: 45847)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Regression Coefficients:

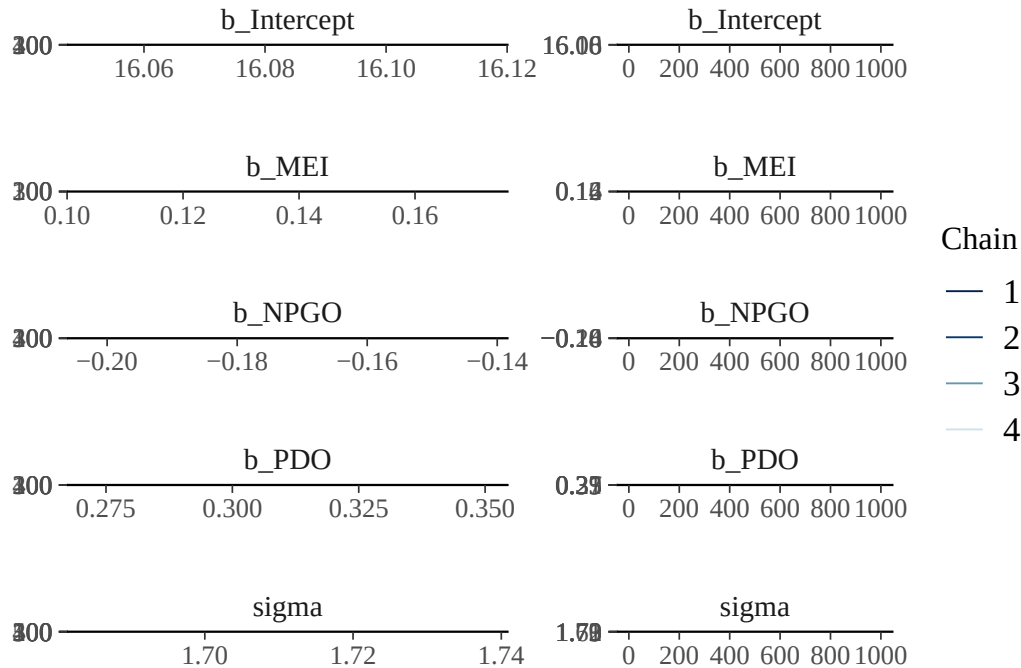
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	16.08	0.01	16.07	16.10	1.00	4388	2777
MEI	0.14	0.01	0.12	0.16	1.00	4031	3089
NPGO	-0.18	0.01	-0.19	-0.16	1.00	4004	3017
PDO	0.32	0.01	0.30	0.34	1.00	3607	3140

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	1.71	0.01	1.70	1.72	1.00	5067	3241

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(m.temp.MEI.NPGO.PDO)
```



Now we can analyze what factors have a direct impact on surface water temp.

MEI: For every one increase in MEI (unit-less index) the surface water temperature increased by an average of 0.14 °C. The affect was significantly different from zero because the compatibility interval ran from 0.12 to 0.16 °C.

NPGO: For every one increase in NPGO (unit-less index) the surface water temperature decreased by an average of 0.18 °C. The affect was significantly different from zero because the compatibility interval ran from -0.19 to -0.16°C.

PDO: For every one increase in PDO (unit-less index) the surface water temperature increased by an average of 0.32 °C. The affect was significantly different from zero because the compatibility interval ran from 0.30 to 0.34 °C.

Now these points alone do not tell us what impact these have on kelp biomass, but they do show us the direct effect that they have on surface water temperature. This is useful because we will be using surface water temperature as a part of out kelp biomass model.

What variables are direct causes of kelp biomass according to our DAG?

Here we are finally analyzing what variables have a direct affect on kelp biomass. Included in the model is nitrate concentration, surface water temperature, and wave height. With our

earlier ggplots early we learned that nitrate concentrations have an approximately logarithmic impact on nitrate concentration. Thus, we will be using a logarithmic model for nitrate in this model. With our earlier ggplots we learned that waves had a nonlinear pattern with a sweet spot for an increase in biomass. Thus, we will be using a non linear model with a k value of 8 to ensure that the line curves enough but there is not an over-fitting of the data.

```
#Including temp + nitrate + waves
#finding the direct effect of mean surface temp, nitrate concentration, and wave height on B
m.biomass.temp.nitrate.waves <-
  brm(data = combined_sitespecific_climind_temp, # Give the model the combined data
      # Choose a gaussian (normal) distribution
      family = gaussian,
      # Specify the model here.
      kelp ~ log10(no3) + mean_surf_temp + s(waves, k = 8),
      # Here's where you specify parameters for executing the Markov chains
      # We're using similar to the defaults, except we set cores to 4 so the analysis runs f
      iter = 2000, warmup = 1000, chains = 4, cores = 4,
      # Setting the "seed" determines which random numbers will get sampled.
      # In this case, it makes the randomness of the Markov chain runs reproducible
      # (so that both of us get the exact same results when running the model)
      seed = 4,
      # Save the fitted model object as output - helpful for reloading in the output later
      file = "outputs/m.biomass.temp.nitrate.waves")

summary(m.biomass.temp.nitrate.waves)
```

```
Family: gaussian
Links: mu = identity
Formula: kelp ~ mean_surf_temp + no3 + waves
Data: combined_sitespecific_climind_temp (Number of observations: 45847)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
      total post-warmup draws = 4000
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	-724911.65	18207.93	-760882.82	-689667.23	1.00	3690
mean_surf_temp	53407.05	1013.48	51421.95	55421.69	1.00	3916
no3	11467.32	407.44	10665.13	12289.97	1.00	3809
waves	-10937.41	1280.11	-13407.28	-8436.01	1.00	3643
Tail_ESS						
Intercept	2648					
mean_surf_temp	2738					

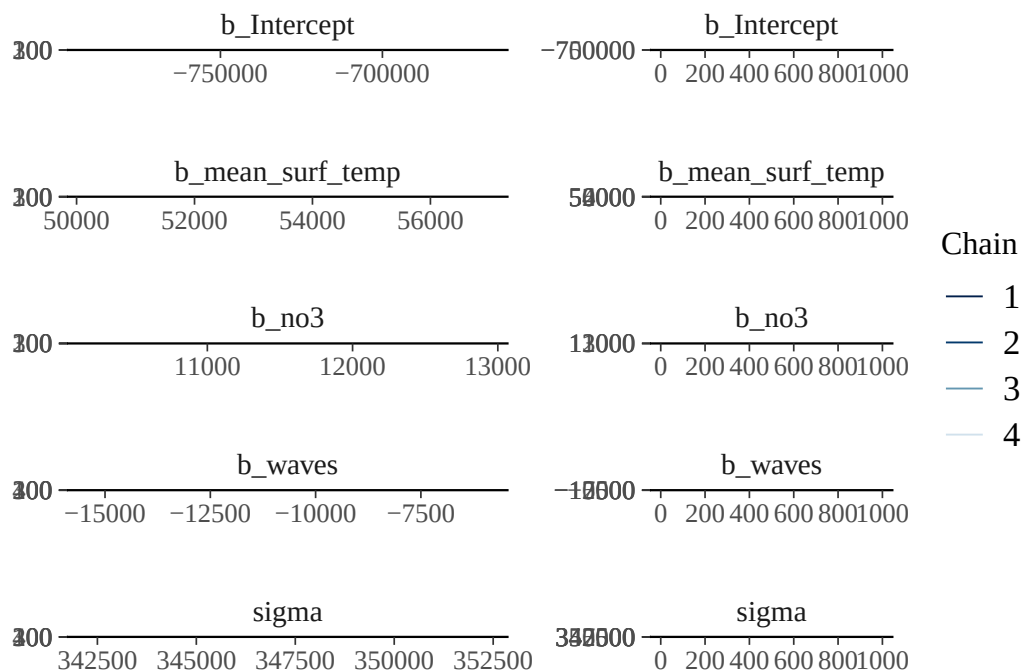
```
no3          2695
waves        2799
```

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	347253.35	1160.36	345031.41	349558.42	1.00	5617	3090

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(m.biomass.temp.nitrate.waves)
```



Now we can analyze what factors have a direct impact on kelp biomass.

Surface Water Temperature: For every 1 °C increase in surface water temperature the kelp biomass increase by an average of about 53000 kg. The affect was significantly different from zero because the comatibility interval ran from about 51000kg to 55000kg.

Nitrate Concentration: For every one increase in micro-moles of nitrate per liter the kelp biomass increased by an average of about 11000kg. The affect was significantly different from zero because the comatibility interval ran from about 10000kg to 12000kg.

Wave Height: For every one meter increase in the height of waves the kelp biomass decreased by an average of about 11000 kg. The affect was significantly different from zero because the comatibility interval ran from about -13000kg to -8000kg.

What are the total effects of NPGO, MEI, and PDO on Kelp biomass

Now we will run a model to see the total affect NPGO, MEI, and PDO has on kelp biomass. In this model we will only be including the NPGO, MEI, and PDO variables. When looking at the dag we can see that surface water temp, nitrate concentration, and wave height are all mediators. Thus, to see the total affect of these the mediators will be left out.

```
#Total effect of NPGO on kelp biomass
m.kelp.NPGO.MEI.PDO.total <-
  brm(
    data = combined_sitespecific_climind_temp, #gives data
    family = gaussian, #sets family as gaussian distribution
    kelp ~ NPGO + MEI + PDO, #Sets parameters based on dag
    iter = 2000, warmup = 1000, chains = 4, cores = 4,
    seed = 4,
    file = "outputs/m.kelp.NPGO.MEI.PDO.total"
  )

summary(m.kelp.NPGO.MEI.PDO.total)
```

```
Family: gaussian
Links: mu = identity
Formula: kelp ~ NPGO + MEI + PDO
Data: combined_sitespecific_climind_temp (Number of observations: 47652)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Regression Coefficients:

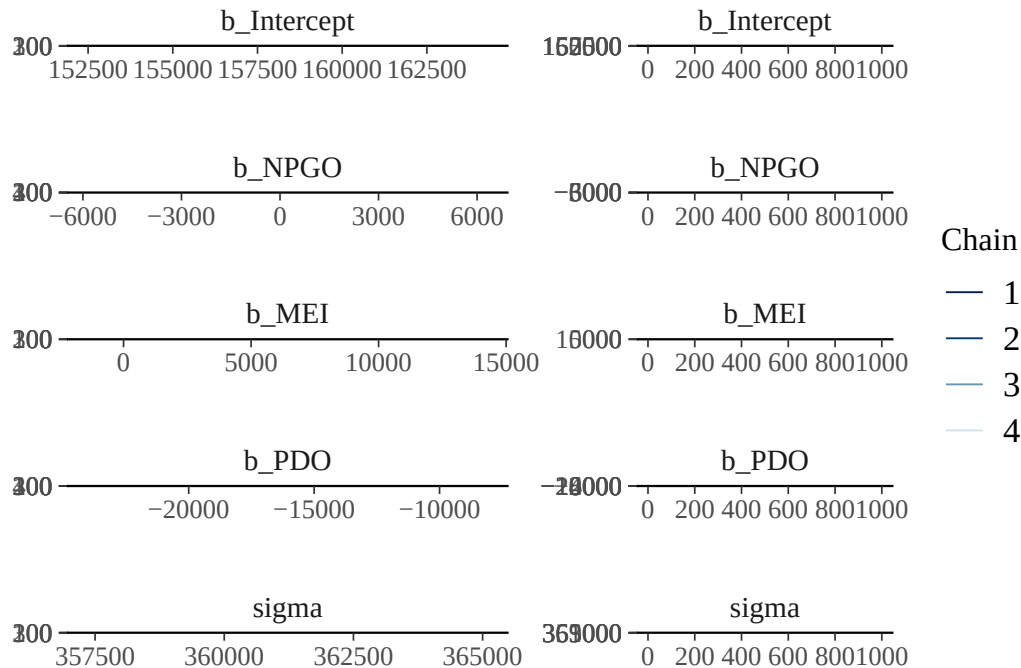
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	158760.84	1710.25	155445.92	162195.73	1.00	4103	2479
NPGO	424.07	1524.41	-2595.04	3295.24	1.00	3781	2830
MEI	6360.01	2243.27	1938.05	10837.76	1.00	3376	2627
PDO	-15713.89	2244.52	-20146.00	-11412.53	1.00	3332	2673

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	361271.10	1189.48	358933.08	363531.21	1.00	5252	3038

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(m.kelp.NPGO.MEI.PDO.total)
```



Now we see what the total effect of our variables are.

NPGO: NPGO did not have an effect which was significantly different from zero. The compatibility interval ran from about -2500kg to 3300 kg.

MEI: For every one increase in MEI (unit-less index) the kelp biomass increased by an average of 6400kg. The affect was significantly different from zero because the compatibility interval ran from 1900kg to 1100kg.

PDO: For every one increase in MEI (unit-less index) the kelp biomass decreased by an average of about 16000kg. The affect was significantly different from zero because the compatibility interval ran from -20000kg to -11000kg.

Conclusion

Kelp biomass can be influenced from a variety, but here we have learned a good bit about some variables impact on biomass. Through our data exploration we were able to create models

which gave us some strong relationships. Through our models and plots we learned that an increase in MEI correlates to an increase in kelp biomass, while an increase in PDO correlates to a decrease in kelp biomass. However, NPGO does not have an effect on kelp biomass which is significantly different from zero. We learned about the direct effects of water temperature, wave height, and nitrate concentrations on kelp biomass as well. We learned that temperature correlated to a significant increase in kelp biomass. However, this trend would likely not continue forever as kelp would certainly die of at extreme temperature. Nitrate correlated to an increase in kelp biomass as well. This is likely due to an increase in nitrogen fixation rates for the algae, which is crucial in their growth and survival. Wave height seemed to have an interesting impact on biomass. Our models said that wave height correlated to a decrease in kelp biomass. However, it crucial to note that in our original ggplot we can see that there is certainly a sweet spot for kelp biomass at different wave heights. Thus, I believe that the wave height variable must be taken with a grain of salt. When all of the trends learned in the data are placed together we can get a pretty solid understanding of what influences kelp biomass. However, there is certainly more research to be done. If more data was collected, for example in our unobserved variables of grazing rates and eutrophication, we could learn much more about what influences kelp biomass. Finally, with that data new conservation plans can be put forth with higher confidence.